Assignment Kit for
Size Counting Standard

Personal Software Process for Engineers: Part I

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Assignment Kit for the Size Counting Standard

Overview

Overview

This assignment kit covers the following topics.

Prerequisites

Prerequisite reading

- Chapter 3

Objectives

The objectives of Size Counting standard are to

- define the size counting standards that are appropriate for the programming language and environment that you will use
- provide a basis for developing a coding standard
- prepare for developing a program to count program size

Size counting standard requirements

Size counting standard requirements

Produce and document a standard for counting program size for the language and environment that you will use in this course.

Submit the completed standard using the format in the template on page 11 with your program 2 assignment package.

Example 1: Pascal size counting standard

Overview

The template is a simplified version of the SEI measurement framework.
- Use this template to describe important items.
- Tailor it to fit your needs or language.

We'll walk through two example size counting standards. The first example is for logical LOC for Pascal programs.

Completing the header

Complete the header as follows:
- the name you give this standard
- the language you are using
- your name
- the date you produced this standard (or revision)

Count type

Choices are logical and physical LOC.
- Logical LOC counts language elements.
- Physical LOC counts text lines.

For this counting standard, you are counting logical LOC.

Statement type

Use this section to define how you will count various types of statements. Consider the following:
- How are you going to count procedure declarations and function prototypes?
- How will you count compiler directives?
- Will you count blank lines or comments? Why or why not?

Clarifications

A fully operational standard generally requires many notes and comments. Use the clarification section for this purpose.

Continued on next page

Pascal LOC Counting Standard Template

Definition Name:	Busqueda	Language:	*java*
Author:	*Bridget Mendez Arano*	Date:	*09-12-25*

Count Type	Type	Comments
Physical/Logical	*Logical*	Se cuentan **líneas lógicas de código (Logical LOC)**. Una Logical LOC es una **unidad de código Java significativa** (sentencia ejecutable, declaración, directiva), independientemente de cuántas líneas físicas ocupe en el archivo de texto.
Statement Type	**Included**	**Comments**
Executable	*yes*	Se cuenta 1 LOC lógica por cada sentencia ejecutable Java, por ejemplo: • Asignaciones (x = 0;) • Llamadas a métodos (simpson.integrateWithDebug();) • Sentencias de control (if, for, while, do, switch, try, catch, finally, return, throw, break, continue) • Sentencias vacías válidas (por ejemplo, el ; vacío dentro de un for).
Nonexecutable:		
Declarations	*yes*	Se cuenta **1 LOC lógica por cada declaración** de: • clase, interfaz o enum (`class Logic { ... }`) • método o constructor • atributo de clase o instancia • variable local o parámetro • Si en una misma línea se declaran **múltiples variables** separadas por comas, se cuenta **1 LOC por cada variable**.

Compiler Directives	*yes*	`package ...;` cuenta como **1 LOC**. • Cada instrucción `import ...;` cuenta como **1 LOC**. • Las anotaciones que aparezcan **solas en una línea** (por ejemplo `@Override` encima de un método) cuentan como **1 LOC no ejecutable**. • No hay directivas de preprocesador tipo `#define` en Java estándar.
Comments	*no*	No se cuentan líneas que sólo contengan comentarios, ni bloques `/* ... */`. • Comentarios al final de una línea de código no agregan LOC adicionales; solo cuenta la sentencia Java.
Blank lines	*no*	Las líneas completamente en blanco no se cuentan.
Other elements		Las líneas que contienen **solo llaves** `{` o `}` **no se cuentan** como LOC por sí mismas. • Los paréntesis, comas, etc. solo se cuentan como parte de las declaraciones o sentencias ya consideradas.
Clarifications		**Examples/Cases**
Note 1		
Note 2		
Note 3		

Example 2: C++ LOC counting standard

How many LOC?

Continued on next page

Example C++ Size Counting Standard

Definition Name:	*Example C++ LOC std.*	Language:	*C++*
Author:	*W.S. Humphrey*	Date:	*12/20/93*

Count Type	**Type**	**Comments**
Physical/Logical	*Logical*	
Statement Type	**Included**	**Comments**
Executable	*Yes*	
Nonexecutable:		
Declarations	*Yes,Notes 3, 4*	
Compiler Directives	*Yes, Note 4*	
Comments	*No*	
Blank lines	*No*	
Other elements		
Clarifications		**Examples/Cases**
Empty statements	*yes*	*";;", lone ;'s, etc.*
Begin...end	*note 1*	
Expression evaluation	*yes*	*when used as sub program arguments*
End symbols	*notes 1,2*	*for terminating executable statements, declarations, bodies*
Then, else, otherwise	*note 1*	
Elseif	*yes*	
Keywords	*yes*	*procedure division, interface, implementation*
Labels	*yes*	*branch destinations when on separate lines*
Note 1		*Count once every occurrence of the following key words: CASE, DO, ELSE, ENUM, FOR, IF, PRIVATE, PUBLIC, STRUCT, SWITCH, UNION, WHILE*
Note 2		*count once every occurrence of the following: ; , {} or };*
Note 3		*count each variable or parameter declaration*
Note 4		*count once each #define, #ifdef, #include, etc. statement*

Example 3: another size counting standard

Overview

For this example standard, the class will select from among the following product categories.

- documents
- interface screens and forms
- database program elements
- maintenance fixes
- any other requested category

We'll then walk through developing the selected size counting standard.

Completing the standard

We'll then walk through the completion of the standard as a class exercise.

Continued on next page

Size Counting Standard Template

Definition Name: ______________________ Language: ______________
Author: ______________________ Date: ______________

Count Type	**Type**	**Comments**
Physical/Logical		
Statement Type	**Included**	**Comments**
Executable		
Nonexecutable:		
Declarations		
Compiler Directives		
Comments		
Blank lines		
Other elements		
Clarifications		**Examples/Cases**
Note		
Note		
Note		
Note		

Evaluation criteria and suggestions

Evaluation criteria	Your standard must be • complete • legible
Suggestions	Keep your standards simple and short. Do not hesitate to copy or build on the PSP materials.

Size Counting Standard Template

Definition Name: ______________________ Language: ____________

Author: ______________________ Date: ____________

Count Type	**Type**	**Comments**
Physical/Logical		
Statement Type	**Included**	**Comments**
Executable		
Nonexecutable:		
Declarations		
Compiler Directives		
Comments		
Blank lines		
Other elements		
Clarifications		**Examples/Cases**
Note		
Note		
Note		
Note		

